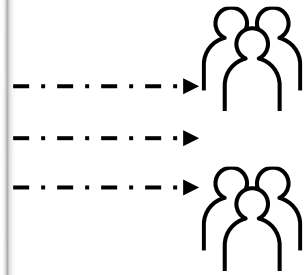
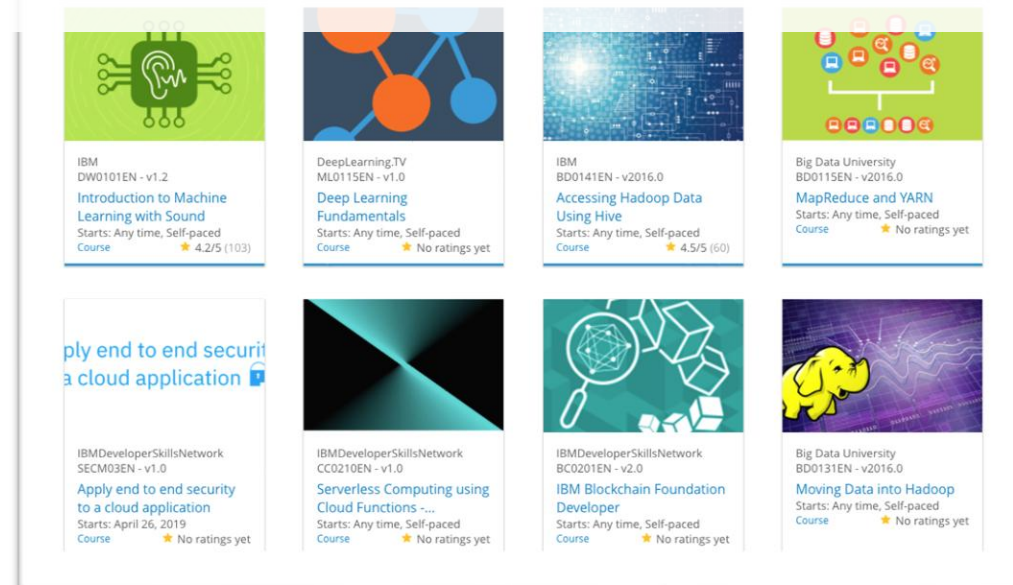


Build a Personalized Online Course Recommender System with Machine Learning

Dhanush V. Hegde
08-10-2026



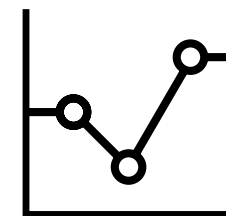
Outline

- Introduction and Background
- Exploratory Data Analysis
- Content-based Recommender System using Unsupervised Learning
- Collaborative-filtering based Recommender System using Supervised learning
- Conclusion
- Appendix

Introduction

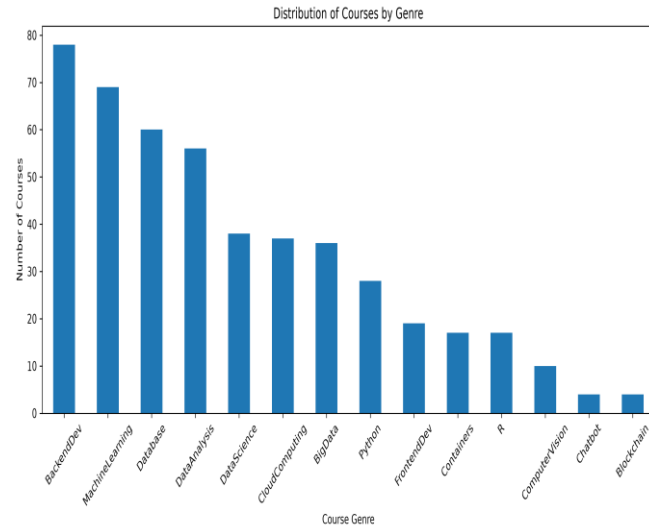
- **Project Background and Context**
- Online learning platforms offer a large number of courses across different learning areas.
- Learners may find it difficult to identify courses that match their interests and previous learning behavior.
- A machine learning-based recommendation system can help learners discover relevant courses.
- This project develops and compares multiple recommendation approaches using course information and learner-course interactions.
- **Hypotheses**
- **H1:** Course content and genre information can be used to generate relevant course recommendations.
- **H2:** Learner interaction history can improve personalized recommendations through collaborative filtering.
- **H3:** Combining different recommendation approaches can provide a more effective recommendation strategy.

Exploratory Data Analysis



Course counts per genre

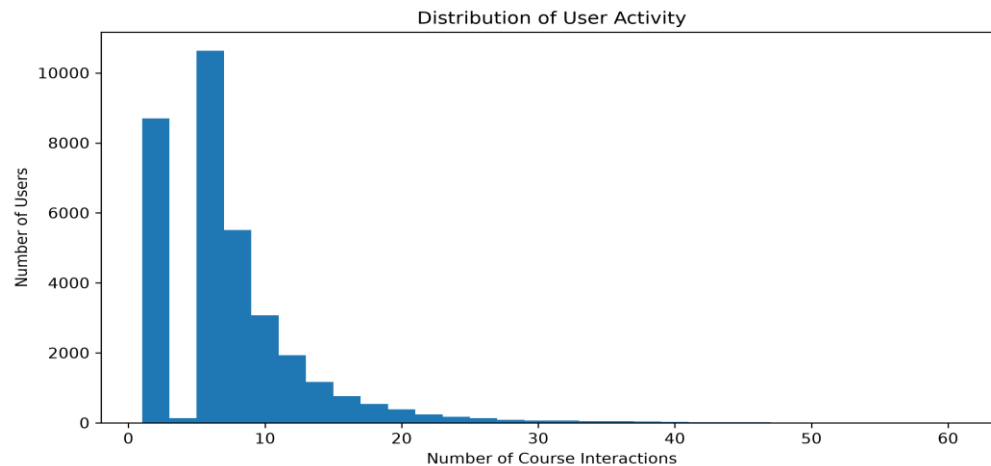
The chart shows the distribution of courses across the different course genres. It highlights which learning areas have the largest representation in the course catalog and which genres have fewer courses.



Course enrollment distribution

The histogram shows the distribution of course interactions across learners. It helps identify how frequently learners interact with courses and whether the dataset contains mostly low-activity or highly active learners.

- A sample histogram may look like the following:



20 most popular courses

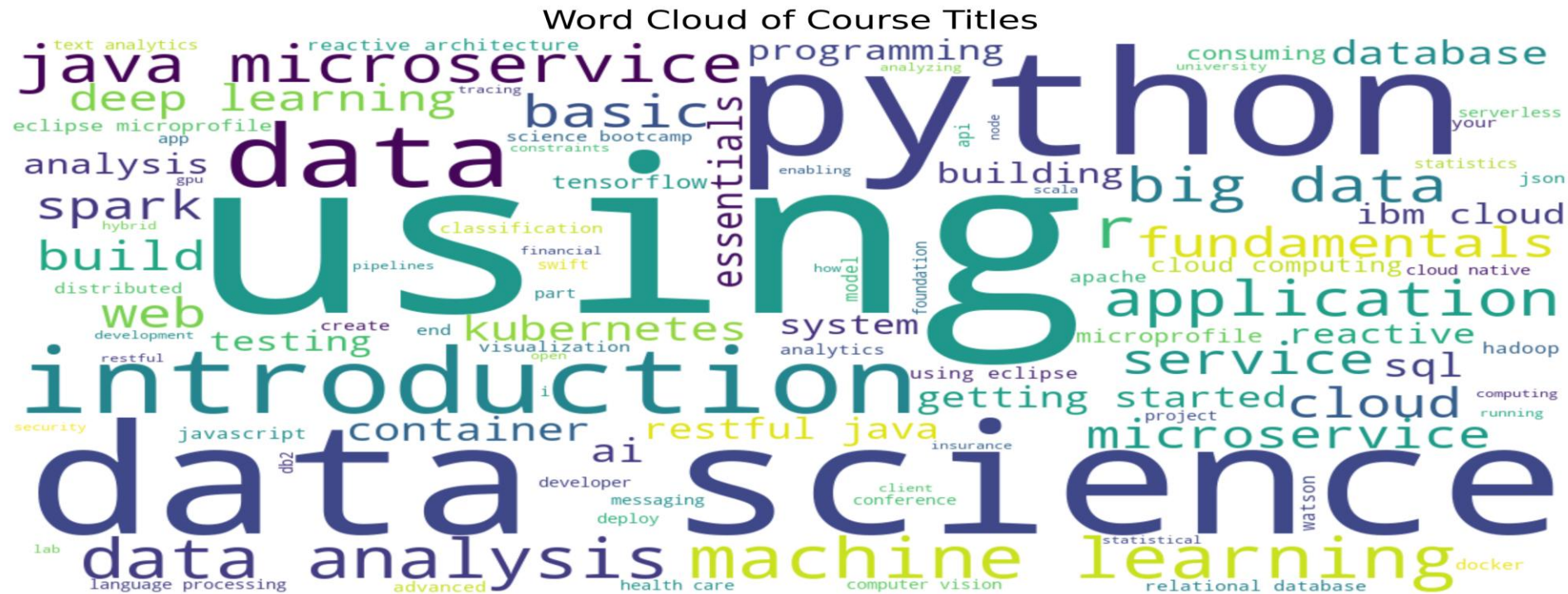
The chart shows the 20 courses with the highest number of learner interactions. It highlights the most popular courses in the dataset based on learner activity.

20 Most Popular Courses

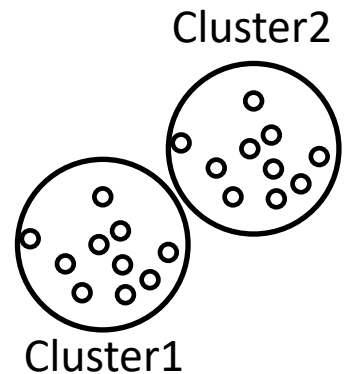
1. python for data science (14936 interactions)
2. introduction to data science (14477 interactions)
3. big data 101 (13291 interactions)
4. hadoop 101 (10599 interactions)
5. data analysis with python (8303 interactions)
6. data science methodology (7719 interactions)
7. machine learning with python (7644 interactions)
8. spark fundamentals i (7551 interactions)
9. data science hands on with open source tools (7199 interactions)
10. blockchain essentials (6719 interactions)
11. data visualization with python (6709 interactions)
12. deep learning 101 (6323 interactions)
13. build your own chatbot (5512 interactions)
14. r for data science (5237 interactions)
15. statistics 101 (5015 interactions)
16. introduction to cloud (4983 interactions)
17. docker essentials a developer introduction (4480 interactions)
18. sql and relational databases 101 (3697 interactions)
19. mapreduce and yarn (3670 interactions)
20. data privacy fundamentals (3624 interactions)

Word cloud of course titles

Explanation: The word cloud summarizes the main terms appearing in the course titles. Larger words represent terms that occur more frequently, helping us identify the major subject areas and topics represented in the course catalog. This analysis also provides an initial understanding of the textual information that can later be used for content-based recommendation.



Content-based Recommender System using Unsupervised Learning



Flowchart of content-based recommender system using user profile and course genres

The system combines learner interaction history with course genre information to create user profile vectors. Course genre vectors are then compared with the learner profile using cosine similarity. Courses with the highest similarity scores are ranked and recommended to the learner.



Evaluation results of user profile-based recommender system

- **Hyper-parameters**
- Number of recommendations: **10**
- Similarity measure: **Cosine Similarity**
- Test users: **50**

On average, how many new/unseen courses have been recommended per user (in the test user dataset)

Average unseen courses per user:
8.34

getting started with the data apache spark makers build (27)

foundations for big data analysis with sql (27)

analyzing big data with sql (27)

spark overview for scala analytics (27)

spark fundamentals ii (27)

spark fundamentals i (27)

analyzing big data in r using apache spark (19)

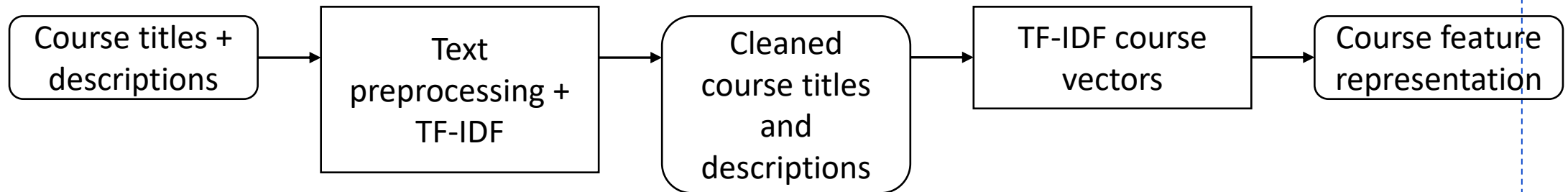
accessing hadoop data using hive (19)

using hbase for real time access to your big data (16)

big data capstone project (13)

Flowchart of content-based recommender system using course similarity

The course similarity recommender uses cleaned course text to create TF-IDF feature vectors. Cosine similarity is then used to compare courses and identify the most similar courses for recommendation.



Evaluation results of course similarity based recommender system

Number of recommendations: 5

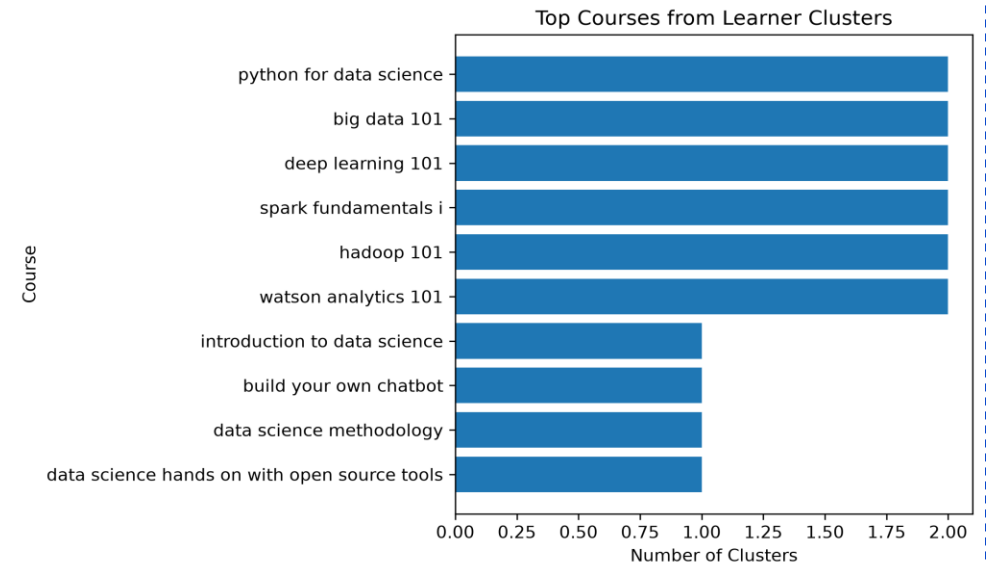
Similarity method: Cosine similarity

Text representation: TF-IDF

Maximum TF-IDF Features: 1000

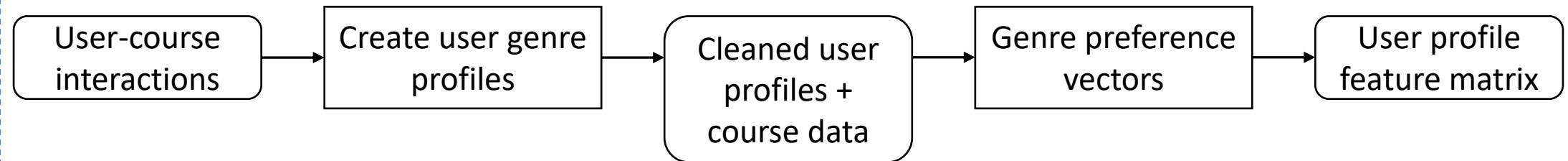
On average, how many new/unseen courses have been recommended per user (in the test user dataset)

Average recommended courses per course: 5.8



Flowchart of clustering-based recommender system

Learner interaction data is processed to create user profiles based on course genre preferences. These profile vectors are then used as features for K-Means clustering to group learners with similar interests. The resulting learner groups can be used to identify courses that are relevant to each cluster.



Evaluation results of clustering-based recommender system

Algorithm: K-Means

Number of clusters: 3

Evaluation Samples: 300 users

Random state:42

On average, how many new/unseen courses have been recommended per user (in the test user dataset)

Average unseen courses per user: 3.53

python for data science (2 cluster(s))

big data 101 (2 cluster(s))

deep learning 101 (2 cluster(s))

spark fundamentals i (2 cluster(s))

hadoop 101 (2 cluster(s))

watson analytics 101 (2 cluster(s))

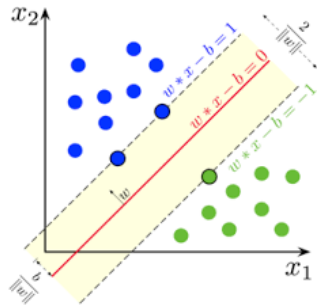
introduction to data science (1 cluster(s))

build your own chatbot (1 cluster(s))

data science methodology (1 cluster(s))

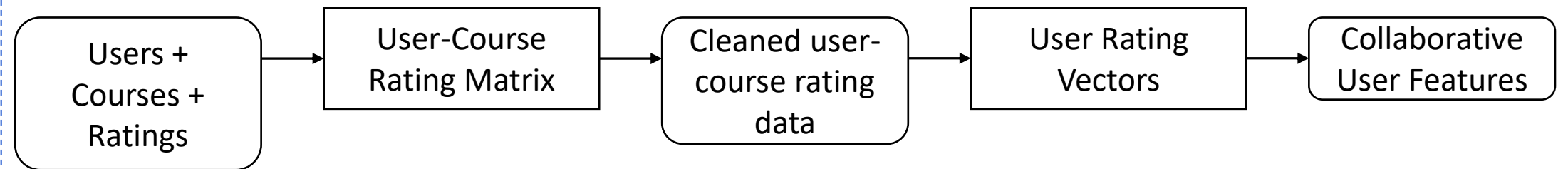
data science hands on with open source tools (1 cluster(s))

Collaborative-filtering Recommender System using Supervised Learning



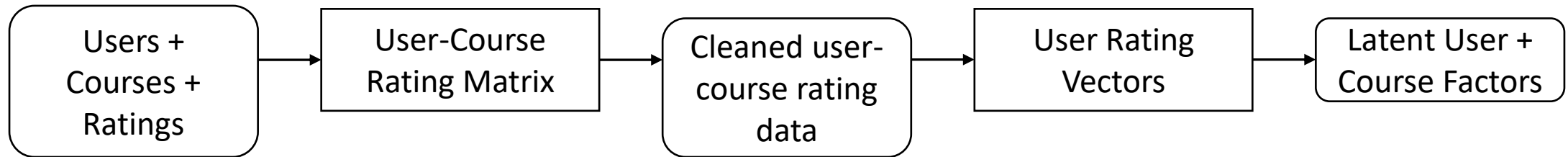
Flowchart of KNN based recommender system

KNN identifies learners with similar course-rating patterns and uses their preferences to recommend courses.



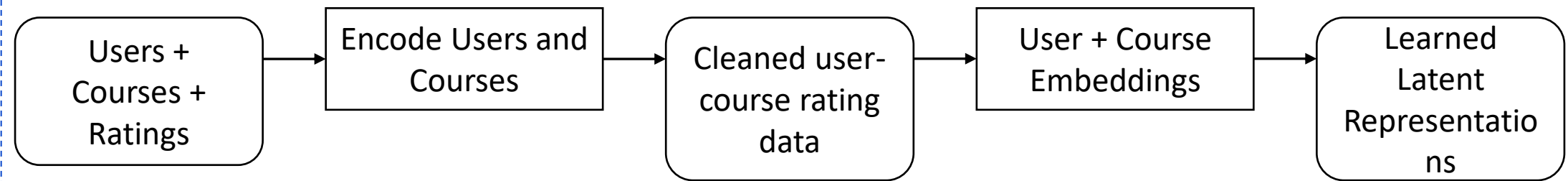
Flowchart of NMF based recommender system

Non-Negative Matrix Factorization (NMF) decomposes the user-course rating matrix into two smaller matrices representing **user preferences** and **course characteristics**. These latent factors are then combined to predict ratings and identify courses that may be relevant to each learner.



Flowchart of Neural Network Embedding based recommender system

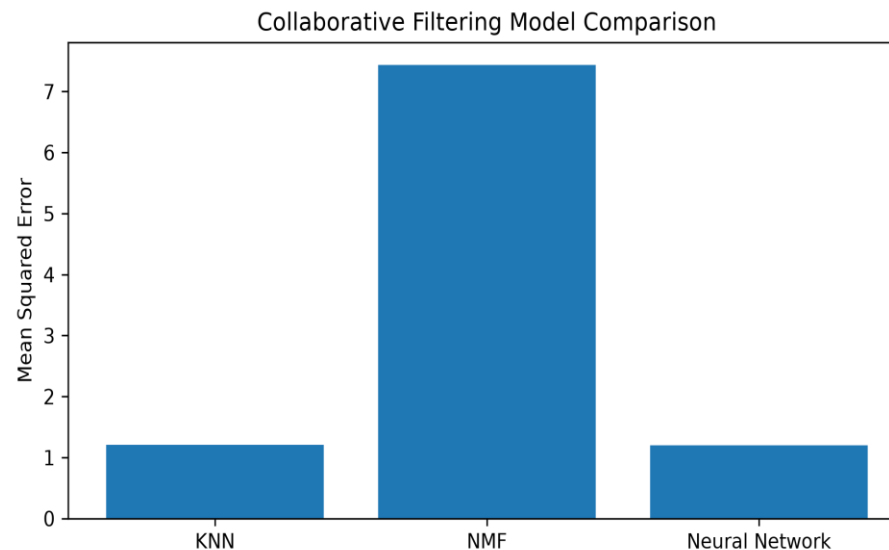
Neural Network Collaborative Filtering learns relationships between users and courses using embedding layers. The model learns latent representations of users and courses and uses them to predict the rating a learner may give to a course.



Compare the performance of collaborative-filtering models

Mean Squared Error (MSE) is used to compare the prediction accuracy of the collaborative-filtering models. A lower MSE indicates that the predicted ratings are closer to the actual ratings.

The comparison helps identify which collaborative-filtering approach provides the most accurate rating predictions on the evaluated sample.



Conclusions

The project developed and compared multiple machine learning approaches for personalized online course recommendation.

- Content-based methods can recommend courses using course content and genre information.
- Collaborative filtering can use learner-course interactions to identify relevant recommendations.
- KNN, NMF, and Neural Networks provide different approaches to learning learner preferences.
- A hybrid recommendation strategy can combine content-based and collaborative signals for better personalization.

Appendix

- Dataset**

- Course genre data
- Processed course data
- User-course ratings

- Machine Learning Methods**

- TF-IDF & Cosine Similarity
- K-Means & PCA
- KNN & NMF
- Neural Network
- MSE comparison
- Recommendation diversity

- Evaluation**

- Catalog coverage

- Implementation**

- Python
- Scikit-learn
- TensorFlow / Keras